FERNADO GEORGE ANAK MANI

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PROFESSIONAL SUMMARY

Data Science graduate from Universiti Malaysia Sabah (CGPA 3.78/4.00, Dean's List) with production experience building computer vision, NLP, and LLM/RAG pipelines during a six-month AI/ML engineering internship.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, JavaScript, HTML, CSS
AI, ML & Data: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG, Feature Engineering, Data Analysis, Power BI
Frameworks & Tools: PyTorch, TensorFlow, OpenAI API, Gemini API, Ollama, FAISS, BM25, Gradio, OpenCV, EasyOCR, LangGraph, CrewAI, MLflow
APIs, Web & Testing: FastAPI, React, Vite, Node.js, Tailwind CSS, pytest, Postman, Git, GitHub Actions (CI/CD)
Databases & Cloud: MongoDB, MySQL, SQLite, SQLAlchemy, AWS (ECS, ECR, S3), Docker, Kubernetes (Minikube)

PROFESSIONAL EXPERIENCE

Breakfast Byte Sdn. Bhd. | **AI/ML Development Intern** | Kuching, Sarawak | **Mar 2026 – Aug 2026**

- Contributed to the enterprise AI chatbot platform redesign (KI Konnekt) for a German client, implementing model routing, retrieval pipelines, state management, UI components, regression testing, and technical documentation.
- Architected the computer vision car wash monitoring pipeline, processing 68 video sessions (95,270 frames) into structured vehicle-visit records via session detection, license-plate OCR, and wash-action classification.
- Trained ConvNeXtV2 action classifiers on 17,451 cropped samples, increasing validation macro F1 from 0.48 to 0.52 and eliminating 84% of short false segments via temporal smoothing algorithms.
- Automated end-to-end daily performance reporting on AWS using Docker, Amazon ECR, Amazon ECS, Amazon S3, and GitHub Actions CI/CD to generate client-ready CSV analytical reports.

EDUCATION

Universiti Malaysia Sabah | **BCompSc (Data Science) with Honours** | **2022 – 2026**

- **CGPA:** 3.78 / 4.00 (Dean's List) | Degree conferred: 30 November 2026

Sarawak Matriculation College | **Matriculation Certificate (Science Module II)** | **2021 – 2022**

- **CGPA:** 4.00 / 4.00 | **Best Computer Science Student Award**

SELECTED PROJECTS & ACHIEVEMENTS

Multimodal Emotion Recognition | **Final Year Project** | **Best Research Award 2026** | **2025 – 2026**

- Conducted comparative analysis of contextual and statistical fusion strategies for robust multimodal emotion recognition combining speech, visual, and text features.
- Developed deep learning architectures using WavLM, DINOv2, ModernBERT, and Bi-GRU under Gaussian noise and missing-modality scenarios, attaining a 71.6% F1-score with ensemble learning.

KinaVis – AI Medical Scribe | **Gold Medal – Festival Idea Sabah 2025** | **2025**

- Architected AI clinical documentation system integrating Google Speech-to-Text, Gemini 2.5 Pro, and Retrieval-Augmented Generation (RAG) with all-MiniLM-L6-v2 vector embeddings.
- Demonstrated reduction in documentation time by 3 to 4 hours per day and cut administrative overhead by 30% during prototype trials.

MatchKami – AI School-NGO Matching Platform | **2nd Prize – JomHack Varsity Challenge 2025** | **2025**

- Engineered an AI-powered matching system connecting under-resourced schools with verified NGOs and donors via smart need-matching algorithms and a Telegram chatbot interface.

HydroGrid Sarawak – Renewable Energy & Hydrogen Intelligence Platform | **GitHub** | **2026**

- Engineered end-to-end telemetry platform ingesting live Open-Meteo data into a PostgreSQL star-schema warehouse with DuckDB OLAP layer, n8n orchestration, and MLflow experiment tracking.
- Trained LightGBM inflow forecasting and Isolation Forest anomaly detection models served via FastAPI; built a LangGraph SQL agent and CrewAI strategic panel with a Streamlit SCADA command center, deployed on Kubernetes.

Kenyalang Ledger – Receipt Digitizer & Accounting Workspace | **GitHub** | **2026**

- Solo-built a FastAPI and React purchase-journal workspace tracing receipt fields to pixels via EasyOCR bounding boxes.
- **Found and fixed a tax-basis bug** that broke GST-inclusive reconciliation and corrupted exports; covered by 311 tests.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

PMFKI Student Association | **Head of Academic and Career Affairs** | **2023 – 2024**

- Directed academic workshops and career development initiatives for undergraduate computer science students.

Hackathon Team Lead & Developer | **4+ Hackathons (MarineHack, JomHack)** | **2023 – 2025**

- Led multi-disciplinary teams across 4+ hackathons (48-hour sprints), owning architecture, task delegation, development and pitching.